Q_1 = 40 years

$Q_3 = 3\left(\frac{N+1}{4}\right)^{th}$ item

$= 3\left(\frac{543+1}{4}\right)^{th}$ item

$= 3 \times 136^{th}$ item

$= 408^{th}$ item

Q_3 = 60 years

Quartiles for Continuous series (grouped data)

Step 1 : Find Cumulative frequencies

Step 2 : Find $\left(\frac{N}{4}\right)$

Step 3: Q_1 class is the class interval corresponding to the value of the cumulative frequency just greater than $\left(\frac{N}{4}\right)$

Step 4 : Q3 class is the class interval corresponding to the value of the cumulative frequency just greater than 3 $\left(\frac{N}{4}\right)$

$$Q_1 = l_1 + \frac{\frac{N}{4} - m_1}{f_1} \times c_1 \quad \text{and} \quad Q_3 = l_3 + \frac{3\left(\frac{N}{4}\right) - m_3}{f_3} \times C_3$$

where N=∑f = total of all frequency values

l_1 = lower limit of the first quartile class

f_1 = frequency of the first quartile class

c_1 = width of the first quartile class

m_1 = c.f. preceding the first quartile class

l_3 = lower limit of the 3rd quartile class

f_3 = frequency of the 3rd quartile class

m_3 = c.f. preceding the 3rd quartile class

c_3 = width of the third quartile class

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